FINITE CLOSURES AND GRAPH CERTIFICATES
SUPPLEMENT TO *COUNTEREXAMPLES AND ENUMERATION FOR CONTRACTIBLE AND DISMANTLABLE GRAPHS*

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ABSTRACT. We describe the finite admissible-edge closures associated with the counterexamples in the accompanying article. We prove the stopping rule and the exhaustiveness of the closure algorithm, explain the natural unfoldings of the 51 stuck K_4-free seeds of order 14, and give a presentation of integral first homology from the triangle relations. We also record the 20 strictly 2-dismantlable graphs of order 12 and relate the earlier 15-vertex example to the present labelling.

1. GRAPH CLASSES AND THE FINITE CONSTRUCTIONS

All graphs are finite and simple. The clique complex of a graph G is denoted by $\mathrm{cl}(G)$. We use induced open neighbourhoods $N_G(v)$ and induced common neighbourhoods $N_G(u, v)$. All homology groups below have integer coefficients unless a field is specified.

The class $\mathfrak{I}_S$ of strong I-contractible graphs contains K_1 and does not contain the empty graph. For a graph of order at least two, $G \in \mathfrak{I}_S$ precisely when either

$$N_G(v),\ G - v \in \mathfrak{I}_S \quad \text{for some } v, \qquad \text{or} \qquad N_G(u, v),\ G - uv \in \mathfrak{I}_S \quad \text{for some } uv \in E(G).$$

The definition is well founded because each argument has smaller $|V| + |E|$. A deletion, or its inverse addition, is *admissible* when the tested neighbourhood lies in $\mathfrak{I}_S$. In the larger class $\mathfrak{I}$, additions and deletions are both permitted, and their neighbourhoods are required to lie in $\mathfrak{I}$. Starting from K_1 gives the least class closed under these four operations [1]. Reversing a strong reduction shows $\mathfrak{I}_S \subseteq \mathfrak{I}$.

A graph of order at least two is *stuck* if its clique complex is acyclic over $\mathbb{F}_2$ and no vertex or edge is admissible for deletion. The order restriction excludes K_1. A stuck graph is outside $\mathfrak{I}_S$, since a strong reduction cannot start.

The accompanying article [3] shows that there are exactly 51 stuck K_4-free graphs of order 14, up to isomorphism. Three are I-contractible: G_{14}^a, G_{14}^b, G_{14}^c, with respectively 44, 45 and 45 edges. Each has an admissible one-edge extension in $\mathfrak{I}_S$. The other 48 have $H_1 \cong \mathbb{Z}/3$ and no other reduced homology. The enumeration and reduction lemmas in the article establish the completeness of this seed list. We study the graphs obtained from these seeds by admissible edge additions, allowing edges that create a K_4 unless a restriction is stated.

2. ADMISSIBLE-EDGE CLOSURES

Definition 2.1. For a graph $G \notin \mathfrak{I}_S$, let $\mathcal{N}(G)$ be the set of isomorphism classes of graphs outside $\mathfrak{I}_S$ obtainable from G by a finite, possibly empty, sequence of admissible edge additions. Let $\mathcal{B}(G)$

consist of the classes $[H + uv] \in \mathfrak{I}_S$ for which $[H] \in \mathcal{N}(G)$ and uv is admissible in H. We call $\mathcal{B}(G)$ the *first-strong boundary*. For a set of seeds $\mathcal{S}$, take the unions $\mathcal{N}(\mathcal{S})$ and $\mathcal{B}(\mathcal{S})$.

The boundary records the first entry into the strong class along some path. A class may have several predecessors or paths from a seed.

Lemma 2.2. *If uv is a missing edge of H, then $N_{H+uv}(u, v) = N_H(u, v)$ as induced graphs. If $H \in \mathfrak{I}_S$ and this common neighbourhood belongs to $\mathfrak{I}_S$, then $H + uv \in \mathfrak{I}_S$.*

Proof. Neither endpoint belongs to the common neighbourhood. Adding uv changes no adjacency within it and creates no new common neighbour. Deleting uv from $H + uv$ leaves H, so the edge alternative in the definition of $\mathfrak{I}_S$ applies. □

Corollary 2.3. *Every intermediate graph of an admissible-addition path ending outside $\mathfrak{I}_S$ also lies outside $\mathfrak{I}_S$.*

Proof. Once a path reaches $\mathfrak{I}_S$, Lemma 2.2 applies to every subsequent addition. □

The *pruned closure algorithm* maintains a list of classes already seen and a queue of non-strong classes still to be expanded. Initially both the seen list and the queue contain the distinct non-strong seed classes. On removing a class $[H]$, test every missing edge uv of a representative. If $N_H(u, v) \in \mathfrak{I}_S$, form $H + uv$. If its class is unseen, insert it: queue it when it is non-strong; when it is strong, record it on the boundary and do not expand it. The tests for $\mathfrak{I}_S$ use the full current graph, including every previously added edge.

Theorem 2.4. *For a finite set of seeds, the pruned closure algorithm terminates. Its non-strong output is exactly $\mathcal{N}(\mathcal{S})$, and its strong output is exactly $\mathcal{B}(\mathcal{S})$.*

Proof. On each branch the order stays fixed and the number of edges increases strictly. A branch from an n-vertex graph has at most $\binom{n}{2} - |E(G)|$ steps. There are finitely many graph classes at each seed order.

Each inserted state is reached by tested admissible additions, so every reported class has the asserted origin. Conversely, take a path to a non-strong target. By Corollary 2.3, every state on the path is non-strong. Whenever such a class is first inserted, it is queued and eventually expanded exactly once. An isomorphism $\varphi : H \to H'$ sends $N_H(u, v)$ isomorphically to $N_{H'}(\varphi(u), \varphi(v))$ and extends to $H + uv \cong H' + \varphi(u)\varphi(v)$. Thus identifying isomorphic states preserves every possible successor class. Induction along the path shows that the target is found. Applying the same argument to the last non-strong predecessor proves the assertion about the boundary. □

If only K_4-free targets are wanted, candidates containing a K_4 may also be discarded: subsequent additions cannot remove that clique. The same argument proves exhaustiveness in this restricted universe.

Remark 2.5. For a 14-vertex graph, a common neighbourhood has at most 12 vertices. The article's classification through order 13 implies that $\mathfrak{I}$ and $\mathfrak{I}_S$ agree on all such neighbourhoods; members of $\mathfrak{I}$ are connected. Consequently I-admissible and $\mathfrak{I}_S$-admissible edge additions coincide at order 14. This observation inherits the finite classification's computational dependence. Theorem 2.4 itself needs only the definition of $\mathfrak{I}_S$ and holds at every order.

Lemma 2.6. *An admissible edge addition induces isomorphisms on integral homology in every degree. It preserves membership in $\mathfrak{I}$ in both directions.*

Proof. Let $C = N_H(u, v)$ and $K = \mathrm{cl}(C)$. The added simplices give

$$\mathrm{cl}(H + uv) = \mathrm{cl}(H) \cup (\Delta^{uv} * K), \qquad \mathrm{cl}(H) \cap (\Delta^{uv} * K) = (\partial\Delta^{uv}) * K.$$

Since $C \in \mathfrak{I}_S \subseteq \mathfrak{I}$, the nonempty complex K is contractible [4, 2]. The intersection is the suspension of K. Both it and the added piece are contractible. The reduced Mayer–Vietoris sequence gives the homology isomorphisms induced by inclusion. The addition and reverse deletion are I-transformations, since their common neighbourhood is the same member of $\mathfrak{I}$. □

3. The closure catalogs

Applying Theorem 2.4 to the three contractible seeds gives

Seed	$\lvert\mathcal{N}(G)\rvert$	$\lvert\mathcal{B}(G)\rvert$	K_4-free members of $\mathcal{N}(G)$
G^a_{14}	16	32	4
G^b_{14}	10	16	3
G^c_{14}	5	8	1

These finite computations use the recursion for $\mathfrak{I}_S$ and exact isomorphism tests. The three non-strong closures overlap, and their union has 29 classes. All 29 belong to $\mathfrak{I}$ by Lemma 2.6. Their eight K_4-free members are exactly the graphs in $\mathfrak{I} \setminus \mathfrak{I}_S$ in row 14 of the article's table. The remaining 21 contain a K_4; their simplicial collapsibility is not classified here.

Conjecture 15 of [1] proposed minimum order 15 for a strong graph with an admissible edge whose deletion leaves the strong class. Every class in $\mathcal{B}(G^a_{14})$ refutes this bound: it has 14 vertices and a representative $H + uv \in \mathfrak{I}_S$ whose admissible edge uv can be deleted to give $H \notin \mathfrak{I}_S$. The count 32 describes this boundary, not every graph with such a deletion.

Proposition 3.1 (Computer-assisted finite catalog)**.** *For the set $\mathcal{S}$ of 51 deposited seeds, $\mathcal{N}(\mathcal{S})$ has 2,271 isomorphism classes. Exactly 29 belong to $\mathfrak{I}$, and the other 2,242 have $H_1 \cong \mathbb{Z}/3$. Exactly 277 classes in this union are K_4-free: eight belong to $\mathfrak{I}$ and 269 have first homology $\mathbb{Z}/3$.*

Proof. The complete pruned search on the 51 seeds, followed by exact isomorphism identification, produces 2,271 non-strong classes. The search tests strong membership by the recursive definition, identifying a subproblem by its complete active graph, and identifies isomorphic results by canonical labelling. The finite verifier reconstructs the three positive-seed closures separately. For the full union, it checks the stored paths, nonmembership in $\mathfrak{I}_S$, pairwise nonisomorphism and integral first homology.

The three contractible seeds contribute precisely the 29-class union above. Every other seed has $H_1 \cong \mathbb{Z}/3$. Lemma 2.6 preserves that group along each addition path, so none of those descendants belongs to $\mathfrak{I}$. The two parts are disjoint, with sizes 29 and $2271 - 29 = 2242$. Testing every four-vertex subset gives the K_4-free counts 8 and 269. □

Row 14 contains 633 $\mathbb{F}_2$-acyclic K_4-free graphs outside $\mathfrak{I}_S$, up to isomorphism. Exact comparison places all 277 K_4-free closure classes inside that set, leaving 356 classes outside these closures. One is encoded by `M?B@`jIja{J}jy{]?`. Thus the 2,271-class union does not exhaust even the K_4-free exceptions at order 14. The positive part is complete in the K_4-free universe by the article's

reduction theorem; the 29-class positive union is not claimed to classify the unrestricted order-14 universe.

4. Natural unfoldings of the stuck seeds

For a K_4-free graph, the clique complex has dimension at most two. Let $d(e)$ be the number of triangles containing an edge e. Begin with a separate closed triangle for each triangle of $\mathrm{cl}(G)$. Whenever $d(e) = 2$, glue the two copies of e preserving the endpoint labels. Make no other identifications. The resulting complex is the *natural unfolding* $U(G)$. When every edge and vertex belongs to a triangle, as for all 51 seeds, the labels define a quotient map $U(G) \to \mathrm{cl}(G)$. The construction identifies triangle corners as well as edges. Distinct edges of the unfolding may have the same image labels.

Proposition 4.1. *The triangle list determines the unfolding's components, vertex links and boundary words by finite gluing. If every vertex link is a circle or an interval, the unfolding is a compact surface with boundary. A connected component with Euler characteristic 1 and one boundary component is a disc; one with Euler characteristic 0 and two boundary components is an annulus.*

Proof. Represent a corner by (T, v), where T is a triangle containing v. For an edge uv in exactly two triangles T, T', identify (T, u) with (T', u) and (T, v) with (T', v). The resulting equivalence classes are the quotient vertices. The glued triangle edges are its edges. Joining triangle pairs along the prescribed edges determines the connected components; no other identification joins two components.

Each incident triangle contributes an interval to a vertex link. Glue interval endpoints exactly when the corresponding triangle edges were glued. A circular link gives a disc neighbourhood; an interval link gives a half-disc neighbourhood. This proves the local surface condition. Unglued triangle edges form the boundary; traversing its cycles gives the boundary words in original labels. Cell counts give the Euler characteristic. The classification of compact connected surfaces now gives the two stated types. □

All 51 unfoldings satisfy this surface condition. The cross-tabulation with the homology of the original clique complex is

Integral $H_1(\mathrm{cl}(G))$	Disc	Annulus	Disc ⊔ annulus
0	3	0	0
$\mathbb{Z}/3$	45	2	1

Thus the three contractible seeds have disc unfoldings, and there are 45 further discs with torsion after boundary identification. The 48/2/1 split is reconstructed from the corner gluing, independently of the homology column. Each of the two connected annular unfoldings has boundary components of lengths 4 and 12. In the disconnected unfolding, the annulus has boundary lengths 8 and 12 and the disc has boundary length 4. Table 1 gives all encodings and boundary lengths; the disconnected unfolding is row 51. The data also record every cyclic boundary word and vertex-link certificate.

5. Integral first homology from triangle relations

The finite checks use an integer presentation that applies to every graph. Its output concerns first homology; higher homology requires additional information.

Proposition 5.1. *Orient all edges of* G *from smaller to larger endpoints, and let* F *be a spanning forest of* G. *Put* $Q = E(G) \setminus E(F)$. *Form an integer matrix* M *with rows indexed by* Q *and columns by triangles. The column for* $i < j < k$ *records the coefficients of* $[j,k] - [i,k] + [i,j]$ *on the edges in* Q. *Then* $H_1(\mathrm{cl}(G);\mathbb{Z}) \cong \operatorname{coker} M$. *If the nonzero Smith factors are* $d_1, \dots, d_r$, *this group is*

$$\mathbb{Z}^{|Q|-r} \oplus \bigoplus_{\substack{1 \le i \le r \\ d_i > 1}} \mathbb{Z}/d_i.$$

Proof. An edge outside F and the unique forest path between its endpoints determine a fundamental cycle. These cycles form an integral basis of $\ker \partial_1$. Equivalently, recording coefficients on Q gives an isomorphism $\ker \partial_1 \to \mathbb{Z}^Q$: the remaining coefficients are uniquely determined by the cycle equations on each tree. Isolated vertices contribute no generator. The boundary of each triangle maps to its column of M. First homology is the cycle lattice modulo triangle boundaries. Higher-dimensional simplices do not change this quotient. The Smith normal form gives the displayed decomposition. □

The Smith computation uses unimodular integer row and column operations and freezes a pivot only after it divides the remaining block. For a connected two-dimensional complex with $\chi = 1$ and finite H_1, the group H_2 is free of rank zero by the Euler characteristic. This gives the higher-homology conclusion for the 48 torsion seeds. For their descendants, Lemma 2.6 supplies the conclusion in every dimension, including when added edges create higher cliques.

6. The strictly 2-dismantlable catalog

A nonempty graph is a cone if a vertex is adjacent to every other vertex. Cones form level (-1)-Dis. For $k \ge 0$, a vertex is k-dismantlable if its open neighbourhood belongs to $(k-1)$-Dis; a graph belongs to k-Dis if successive such deletions reduce it to K_1. The empty graph belongs to no level. The hierarchy satisfies 1-Dis $\subseteq$ 2-Dis $\subseteq \mathfrak{I}_{SV}$, where $\mathfrak{I}_{SV}$ is defined by the recursion for $\mathfrak{I}_S$ using only vertex deletions [1].

A graph is *strictly 2-dismantlable* if it belongs to 2-Dis and not to 1-Dis.

Proposition 6.1 (Computer-assisted)**.** *The minimum order of a graph in* 2*-Dis outside* 1*-Dis is 12. There are exactly 20 isomorphism classes of that order.*

Proof. Vertex deletion cannot merge components, and an isolated vertex cannot be deleted because its neighbourhood is empty. Thus every 2-dismantlable graph is connected, so the connected universes in the table suffice. For orders 3 through 11, the article's table gives equal counts for 1-Dis and $\mathfrak{I}_{SV}$. The inclusions force equality with 2-Dis. At orders one and two, its only members are K_1 and K_2, both in 1-Dis. At order 12 the difference is

$$22\,666\,525\,713 - 22\,666\,525\,693 = 20.$$

The 20 deposited graphs are pairwise nonisomorphic. Exact recursion places each in 2-Dis and outside 1-Dis. They exhaust the difference and establish existence at the minimum order. Completeness and minimality inherit the table's enumeration dependence. □

Table 2 records the 20 graphs and their elementary invariants. Automorphism-group orders are exact counts of adjacency-preserving permutations, included as catalog invariants.

7. The earlier 15-vertex example

Let D_{15} have encoding `N|thHsL‘sHoI‘RPOwLG`. It has 15 vertices, 46 edges and 32 triangles. The correspondence with Figure 2A of [1] is

present	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
published	1	4	13	12	11	3	14	15	10	5	2	6	7	8	9

Under this bijection the complete triangle lists agree. The image of the full edge set also agrees with the union of edges in the recorded published triangles. These comparisons identify the recorded triangulation and its graph in the two labellings. The missing edge $\{9, 12\}$ corresponds to the missing edge $\{5, 7\}$ in the published labelling.

The unfolding is a disc with face vector $(23, 54, 32)$ and boundary word

$$(0, 1, 5, 10,\ 0, 1, 5, 10,\ 0, 10, 5, 1).$$

Its image is a four-cycle, traversed twice in one direction and once in the opposite direction. This is the dunce-hat attachment, of degree one. Contracting a spanning tree of that cycle to a point, a homotopy equivalence, gives a circle with a two-cell attached by degree one, so $\mathrm{cl}(D_{15})$ is contractible. The disc verifier checks interior injectivity and that all identifications occur on the boundary.

For a direct graph certificate, add $\{9, 12\}$. The common neighbourhood is strong, and the resulting graph has the strong reduction in Table 3. Every displayed neighbourhood is taken in the current graph; its stored vertex reduction is checked recursively. To see that D_{15} itself is non-strong, inspect its neighbourhoods. Each vertex neighbourhood is triangle-free and is not a tree; each edge's common neighbourhood is edgeless with at least two vertices. A triangle-free graph in $\mathfrak{I}_S$ must be a nonempty tree, since its clique complex is contractible. Thus none of these neighbourhoods belongs to $\mathfrak{I}_S$, and D_{15} is stuck.

8. Data and verification

The encodings, triangle lists, boundary words and reductions are in the companion repository:

https://github.com/Geometric-and-Combinatorial-Structures/minimal-I-contractible-not-strong.

The finite reproduction route described in the repository checks the disc presentations, 51 unfoldings, three small closures and displayed reductions. It also checks the paths and homology groups in the 2,271-class catalog and provides a command to regenerate the full seed-generated union. A separate guide describes the complete enumerations underlying the table and seed classification. The verification map in `docs/verification-scope.md` identifies the finite checks and historical enumeration records supporting each result.

Appendix A. Catalog tables

The machine-readable catalogs are the authoritative source for copying graph6 encodings; PDF text extraction may substitute glyphs.

Table 1: The 51 stuck K_4-free seeds of order 14. Here m counts edges, t triangles, and $a = |\mathrm{Aut}(G)|$. The final column gives all boundary lengths of the natural unfolding. Rows 48 and 49 are annuli; row 51 is a disc together with an annulus; the other rows are discs.

No.	Encoding	m	t	a	H_1	Boundary
1	M?b@bFHNEk@~|k}U?	44	31	1	0	15
2	M?B@thku?~Lylk}[?	45	32	2	0	16
3	M?r@d`ifbiFfo}~o?	45	32	2	0	18
4	M?B@cXamAu\r{]V{?	43	30	2	$\mathbb{Z}/3$	12
5	M?B@cXemAuB}ze{]?	43	30	2	$\mathbb{Z}/3$	12
6	M?BDA_wqpmJvj]|w?	43	30	2	$\mathbb{Z}/3$	12
7	M?`@E_wDf[UnU}z{?	43	30	6	$\mathbb{Z}/3$	12
8	M?`adQWT\LK^du^w?	43	30	2	$\mathbb{Z}/3$	12
9	M?`adQWtVDUr[]^w?	43	30	2	$\mathbb{Z}/3$	12
10	M?bB@_WmbRJZ|M~w?	43	30	1	$\mathbb{Z}/3$	12
11	M?bB@_WsbX^Jlm~w?	43	30	1	$\mathbb{Z}/3$	12
12	M?bB@_WsequuZZ~w?	43	30	1	$\mathbb{Z}/3$	12
13	M?bB@_Wu`ZJjlm~w?	43	30	2	$\mathbb{Z}/3$	12
14	M?bB@_WubXNJlm~w?	43	30	2	$\mathbb{Z}/3$	12
15	M?bB@aWYTkZZ^M|w?	43	30	1	$\mathbb{Z}/3$	12
16	M?bB@aWZSmXZ|wNm?	43	30	1	$\mathbb{Z}/3$	12
17	M?bB@aWZTkFf|wuu?	43	30	1	$\mathbb{Z}/3$	12
18	M?bB@aWdbYFv|wuu?	43	30	1	$\mathbb{Z}/3$	12
19	M?bB@aWdbZEv|wuu?	43	30	1	$\mathbb{Z}/3$	12
20	M?bB@aWqplQzZ]|w?	43	30	1	$\mathbb{Z}/3$	12
21	M?bB@aWyTkYZ^M|w?	43	30	1	$\mathbb{Z}/3$	12
22	M?bB@aXdbY[{Nmuu?	43	30	1	$\mathbb{Z}/3$	12
23	M?bB@aXlbYFfxwuu?	43	30	2	$\mathbb{Z}/3$	12
24	M?bB@aXyTkYZxw^M?	43	30	1	$\mathbb{Z}/3$	12
25	M?otU`SmVEKm]M^o?	43	30	4	$\mathbb{Z}/3$	12
26	M?qa``{rfEEymitY?	43	30	4	$\mathbb{Z}/3$	12
27	M?qa``{ufEUimitY?	43	30	4	$\mathbb{Z}/3$	12
28	M?r@daiRbrTYd{tb?	43	30	1	$\mathbb{Z}/3$	12
29	M?`Db_kVcyl]hyvw?	44	31	2	$\mathbb{Z}/3$	15
30	M?`Db_khi{L\nUvw?	44	31	1	$\mathbb{Z}/3$	15
31	M?`adL[mBkvLY{d{?	44	31	2	$\mathbb{Z}/3$	15
32	M?`cu_whRRFZ~Wtm?	44	31	1	$\mathbb{Z}/3$	15
33	M?`cv@UNDNZMtsY{?	44	31	1	$\mathbb{Z}/3$	15
34	M?b@bCw`usLvvw{]?	44	31	1	$\mathbb{Z}/3$	15
35	M?qa``[NNEYlme~o?	44	31	2	$\mathbb{Z}/3$	15
36	M?qa``[iUEylNm~o?	44	31	1	$\mathbb{Z}/3$	15
37	M?qa``[icjk^Nk~o?	44	31	2	$\mathbb{Z}/3$	15
38	M?qa``[iej[ZNk~o?	44	31	2	$\mathbb{Z}/3$	15
39	M?qa``kLeE^wt]i}?	44	31	2	$\mathbb{Z}/3$	15
40	M?qa``kMmMIniu~o?	44	31	2	$\mathbb{Z}/3$	15

Continued on the next page

No.	Encoding	m	t	a	H_1	Boundary
41	M?qa``kifEfUtZ~o?	44	31	1	$\mathbb{Z}/3$	15
42	M?qa`awkmiNdwu~o?	44	31	2	$\mathbb{Z}/3$	15
43	M?qa`awwRxYj{k~o?	44	31	2	$\mathbb{Z}/3$	15
44	M?qa`g{guLzwmkU]?	44	31	2	$\mathbb{Z}/3$	15
45	M?qa`jSwubncvomd?	44	31	2	$\mathbb{Z}/3$	15
46	M?qaejaLe[ngmh{[?	44	31	2	$\mathbb{Z}/3$	15
47	M?`FSzcVBmVFs{J{?	45	32	4	$\mathbb{Z}/3$	18
48	M?qa``{ws}Lxmkt[?	45	32	8	$\mathbb{Z}/3$	4,12
49	M?qa``{wvkVUt[}Q?	45	32	8	$\mathbb{Z}/3$	4,12
50	MCOfczgVB]UeM{p{?	45	32	4	$\mathbb{Z}/3$	18
51	MCp`faiFf{W}][js?	47	34	2	$\mathbb{Z}/3$	4,8,12

Table 2: The 20 graphs of order 12 in 2-Dis outside 1-Dis. The columns give edges m, triangles t, four-cliques q, clique number ω, degrees δ, Δ, and automorphism-group order a.

No.	Encoding	m	t	q	ω	δ	Δ	a
1	KEhutxmv|~^{	46	69	34	4	6	9	4
2	KEjer^{z}~^{	47	73	37	4	6	9	2
3	KQyurzU~t~^{	47	72	37	5	7	9	4
4	KQyurzU~u~^{	47	73	39	5	6	9	2
5	KEhvtzm||~^{	48	78	45	5	6	9	2
6	KEjeu~m^~z^r	48	77	42	5	6	9	1
7	KEnfbzm||~^{	48	75	38	4	7	9	4
8	KEnfbzm~]~^{	48	75	38	4	7	9	2
9	KEnfbzm~^v^{	48	75	38	4	7	9	2
10	KQyurzuv|~^{	48	77	44	5	7	9	4
11	KQyurzuv~}^{	48	76	42	5	7	9	2
12	KEht~jzz~}^f	49	82	50	5	7	9	2
13	KEjflzz^~}^f	49	81	47	5	7	9	2
14	KEnb~jl~t~^{	49	81	48	5	7	9	2
15	KEnb~jl~v|^{	49	80	45	5	8	9	2
16	KEnff^{^~v^{	49	80	44	5	7	9	1
17	KEnfnZ{~r~^{	49	79	42	5	8	9	4
18	KUZurzm~vv^{	49	81	48	5	8	9	4
19	KEnb~rvz~}^f	50	86	54	5	8	9	2
20	KEzftzv^~}^f	50	85	51	5	8	9	2

Table 3: A strong reduction of $D_{15} + \{9, 12\}$. The symbols v and e indicate vertex and edge deletions. The last column lists vertex deletions in the tested neighbourhood, followed by its surviving vertex in brackets. The main reduction ends at vertex 14.

Step	Delete	Current neighbourhood	Link reduction
1	$v_{\{11\}}$	$\{0,7,9,12\}$	$0,7,9$; [12]
2	$e_{\{0,9\}}$	$\{10,12\}$	10 ; [12]
3	$e_{\{0,12\}}$	$\{10\}$	; [10]
4	$v_{\{10\}}$	$\{0,5,8,9,12,14\}$	$0,9,12,5,8$; [14]
5	$v_{\{8\}}$	$\{4,5,7,14\}$	$5,4,7$; [14]
6	$v_{\{4\}}$	$\{0,1,3,5,7\}$	$5,1,0,3$; [7]
7	$v_{\{0\}}$	$\{1,2,3,14\}$	$3,2,1$; [14]
8	$v_{\{1\}}$	$\{2,5,13,14\}$	$2,5,13$; [14]
9	$v_{\{2\}}$	$\{3,5,6,7\}$	$3,5,6$; [7]
10	$v_{\{3\}}$	$\{7\}$	; [7]
11	$v_{\{5\}}$	$\{6,9,12,13\}$	$6,9,12$; [13]
12	$v_{\{6\}}$	$\{7,9\}$	7 ; [9]
13	$v_{\{7\}}$	$\{9,12,13,14\}$	$9,12,13$; [14]
14	$v_{\{9\}}$	$\{12\}$	; [12]
15	$v_{\{12\}}$	$\{13\}$	; [13]
16	$v_{\{13\}}$	$\{14\}$	; [14]

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